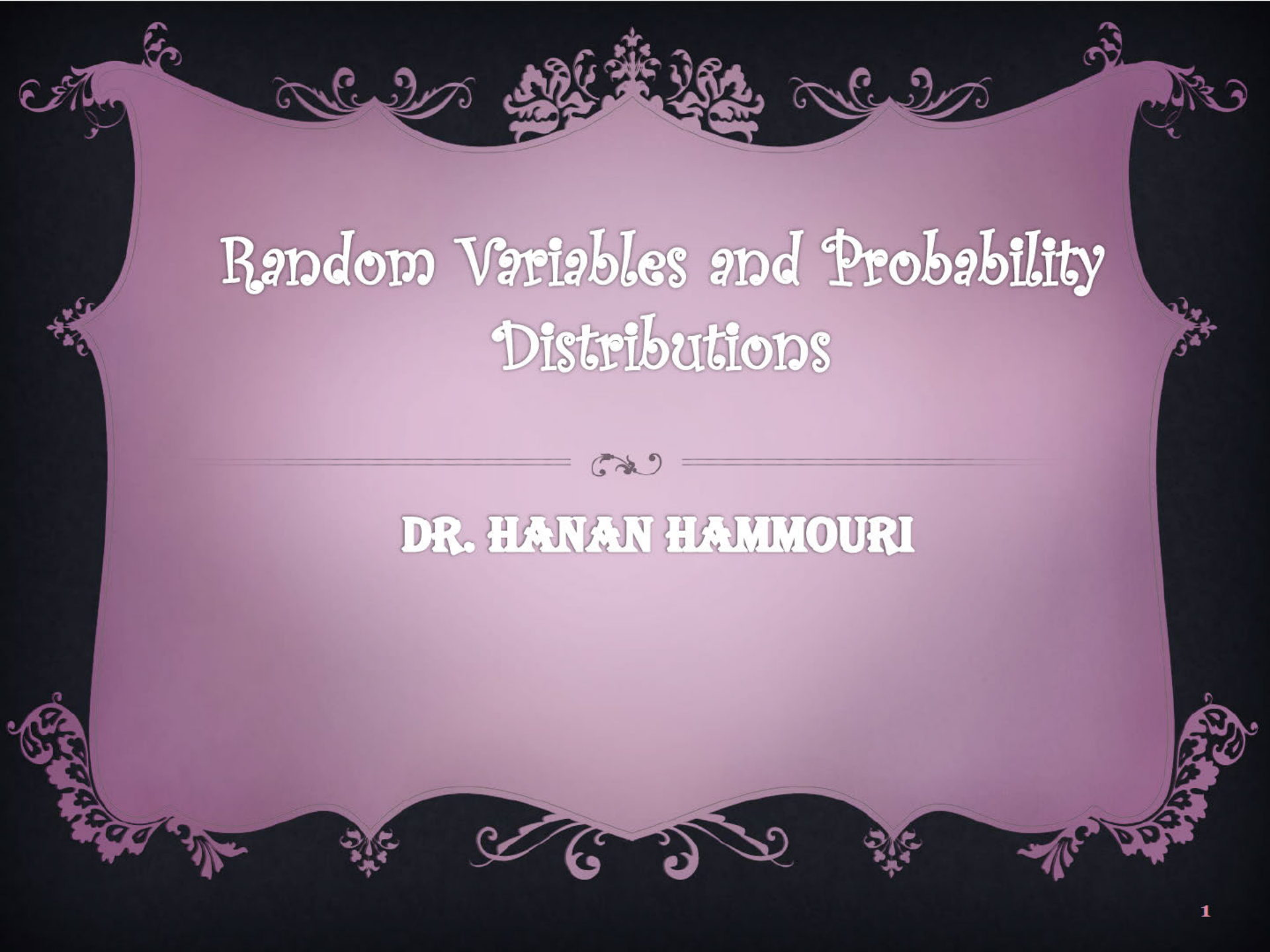




Random Variables and Probability Distributions

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RANDOM VARIABLES

A random variable is a *numerical* measure of the outcome from a probability experiment, so its value is determined by chance. Random variables are denoted using letters such as X (*capital letter*).

Definition

A random variable is a function that maps outcomes of a random experiment to real numbers.

RANDOM VARIABLE

In many situations, we are interested in numbers associated with the outcomes of a random experiment.

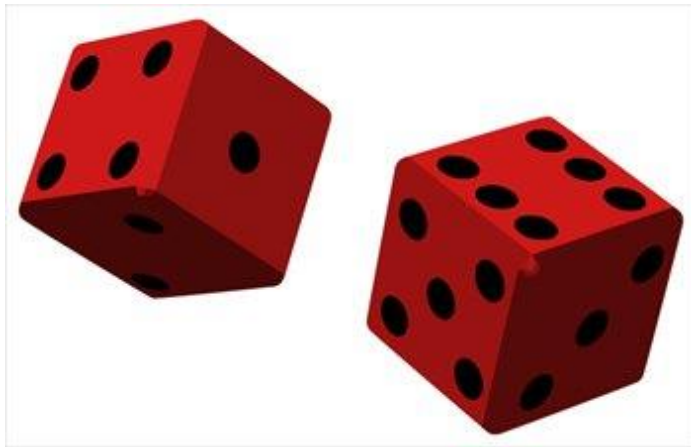
A box of 6 eggs is rejected if it contains one or more broken eggs. If we examine 10 boxes of eggs, we may be interested in

X1 - the number of broken eggs in the 10 boxes

X2 - the number of boxes rejected

ROLLING TWO DICE

EXAMPLE



We can define many random variables

V1: If both faces are the same put 1 if not put 2

V2: the sum of the two faces

V3: the difference between

The two faces

TWO TYPES OF RANDOM VARIABLES

A discrete random variable can assume a countable number of values.

- Number of steps to the top of the Eiffel Tower*

A continuous random variable can assume any value along a given interval of a number line.

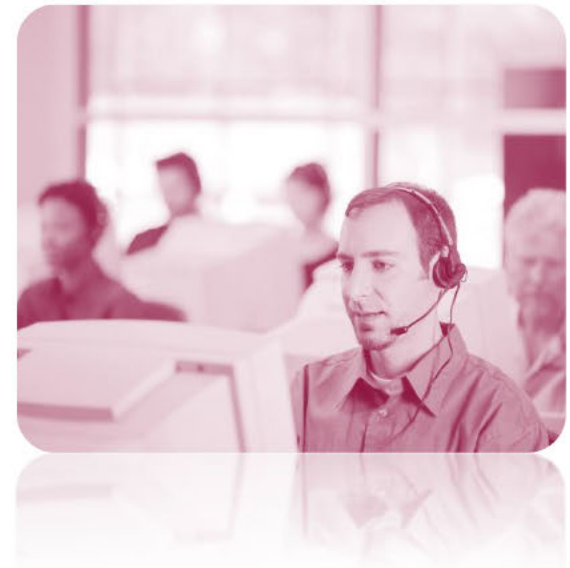
- The time a tourist stays at the top once s/he gets there



TWO TYPES OF RANDOM VARIABLES

Discrete random variables

- Number of sales
- Number of calls
- Shares of stock
- People in line
- Mistakes per page



Continuous random variables

- Length
- Depth
- Volume
- Time
- Weight

PROBABILITY DISTRIBUTIONS FOR DISCRETE RANDOM VARIABLES

The *Probability Mass Function(PMF)* of a discrete random variable is a graph, table or formula that specifies the probability associated with each possible outcome the random variable can assume.

- $P(x) \geq 0$ for all values of x
- $\sum P(x) = 1$
- We find $P(X = x) = f(x)$

EXAMPLES

Ex1: random variable x has these probabilities in the table: Is this table denote PMF??

Yes, all probabilities are > 0 and they sum to 1



x	P(x)
1	.32
2	.23
3	.19
4	.19
5	.07

Ex2: $f(x) = \frac{x}{6}$ where $x = 1, 2, 3$

Is this a PMF??

x	1	2	3
f(x)	1/6	2/6	3/6

Then

Yes, all probabilities are > 0 and they sum to 1

EXAMPLES

Ex: A random variable x has these probabilities in the following table:

Find

$$P(x = 3) = 0.19$$

$$P(x < 5) = 0.93$$

$$P(3 < x < 5) = 0.19$$

$$P(3 \leq x \leq 5) = 0.45$$

$$P(2 < x) = 0.45$$

$$P(0 < x < 6) = 1$$

x	P(x)
1	.32
2	.23
3	.19
4	.19
5	.07

Example: Consider tossing a fair coin 3 times.
Define X = the number of heads obtained.

Value	0	1	2	3
Probability	$1/8$	$3/8$	$3/8$	$1/8$

$X = 0$: TTT

$X = 1$: HTT THT TTH

$X = 2$: HHT HTH THH

$X = 3$: HHH

Q1: What is the probability of tossing at least two heads?

Ans: $P(X \geq 2) = P(X=2) + P(X=3) = 3/8 + 1/8 = 1/2$

Q2: What is the probability of tossing fewer than three heads?

Ans: $P(X < 3) = P(X=0) + P(X=1) + P(X=2) = 1/8 + 3/8 + 3/8 = 7/8$

Or $P(X < 3) = 1 - P(X = 3) = 1 - 1/8 = 7/8$

EXAMPLE

Is the following a probability mass function?

x	$P(X = x)$	
0	0.16	No, because one of the probabilities is negative
1	0.18	
2	0.22	
3	-0.10	
4	0.3	
5	0.04	

EXAMPLE

Is the following a probability mass function?

x	$P(X = x)$
0	0.16
1	0.18
2	0.22
3	0.10
4	0.3

No, because
they don't sum
to 1

EXAMPLE

If $f(x) = cx$ where $x = 0, 1, 2$,

What is the value of c that makes this function a PMF??

X	0	1	2
f(x)	0	c	2c

Then $0+c+2c=1 \longrightarrow 3c=1 \longrightarrow c=1/3$

X	0	1	2
f(x)	0	1/3	2/3

And all the probabilities are none negative

CUMULATIVE DISTRIBUTION FUNCTION (CDF)

Definition

The cumulative distribution function of a random variable X is the function

$$F(x) = P(X \leq x).$$

The cumulative distribution function gives the probability that the variable takes a value less than or equal to x and is defined for all real x .

CDF

The cumulative distribution function of a discrete random variable must satisfy:

- ❶ $F(x) = P(X \leq x) = \sum_{x_i \leq x} f(x_i).$
- ❷ $0 \leq F(x) \leq 1$ for all x .
- ❸ If $x \leq y$ then $F(x) \leq F(y)$.

Example

Suppose the range of a discrete random variable is $\{0, 1, 2, 3, 4\}$ and its probability mass function is $f(x) = \frac{x}{10}$. What is its cumulative distribution function?

EXAMPLE

Example

First of all, for any $x < 1$, $F(x) = \sum_{x_i \leq 0} f(x_i) = f(0) = 0$.

Next, for $1 \leq x < 2$, $F(x) = \sum_{x_i \leq 1} f(x_i) = f(0) + f(1) = \frac{1}{10}$

For $2 \leq x < 3$, $F(x) = f(0) + f(1) + f(2) = \frac{3}{10}$

Continuing in the same way, we find that:

$$F(x) = \begin{cases} 0 & x < 1 \\ \frac{1}{10} & 1 \leq x < 2 \\ \frac{3}{10} & 2 \leq x < 3 \\ \frac{6}{10} & 3 \leq x < 4 \\ 1 & 4 \leq x. \end{cases}$$

EXAMPLE

If we have the following CDF, find the  PMF

$$F(x) = \begin{cases} 0 & x < -15 \\ 0.25 & -15 \leq x < 1 \\ 0.5 & 1 \leq x < 3 \\ 0.6 & 3 \leq x < 13 \\ 1 & 13 \leq x \end{cases}$$

x	-15	1	3	13
f(x)	0.25	0.25	0.1	0.4

PROBABILITY DENSITY FUNCTIONS(PDF)

Definition

For a continuous random variable X , a probability density function is a function such that

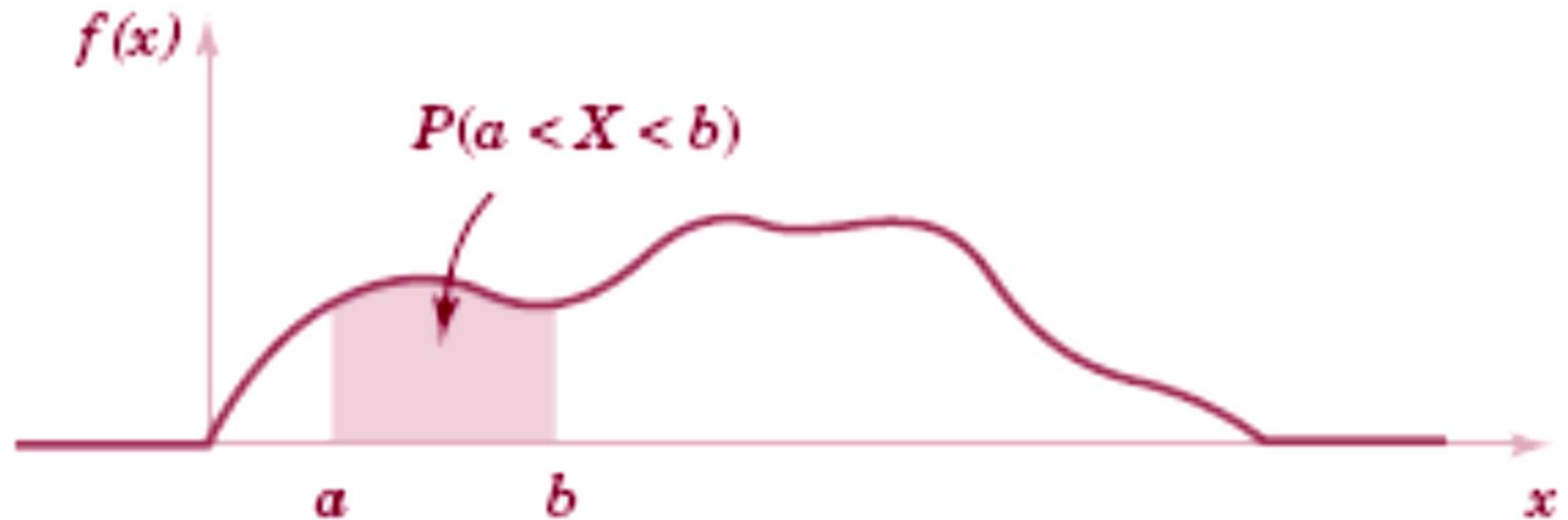
$$(1) \quad f(x) \geq 0$$

$$(2) \quad \int_{-\infty}^{\infty} f(x) dx = 1$$

$$(3) \quad P(a \leq X \leq b) = \int_a^b f(x) dx = \text{area under } f(x) \text{ from } a \text{ to } b$$

for any a and b

PROBABILITY DENSITY FUNCTIONS



Probability determined from the area under $f(x)$.

PROBABILITY DENSITY FUNCTIONS

If X is a continuous random variable, for any x_1 and x_2 ,

$$P(x_1 \leq X \leq x_2) = P(x_1 < X \leq x_2) = P(x_1 \leq X < x_2) = P(x_1 < X < x_2)$$

Because $P(X=x) = 0 \ \forall \ x \in (a, b)$

EXAMPLE

If $f(x) = x/2, x \in (0,2)$

a. Is this a PDF?

Yes, its positive function $f(x) \geq 0$

$$\text{And } \int_0^2 \frac{x}{2} dx = \left. \frac{x^2}{4} \right|_0^2 = 1$$

EXAMPLE

If $f(x) = \frac{x}{2}$, $x \in (0,2)$ Find:

$$a) P(0 < X < 0.5) = \int_0^{0.5} \frac{x}{2} dx = \left. \frac{x^2}{4} \right|_0^{0.5} = 0.0625$$

$$b) P(-1 \leq X \leq 0.5) = P(0 \leq X \leq 0.5) = \int_0^{0.5} \frac{x}{2} dx = \left. \frac{x^2}{4} \right|_0^{0.5} = 0.0625$$

$$c) P(-1 < X < 3) = P(0 \leq X \leq 2) = 1$$

$$d) P(.6 < X < .8) = \int_{0.6}^{0.8} \frac{x}{2} dx = \left. \frac{x^2}{4} \right|_{0.6}^{0.8} = 0.07$$

CDF FOR CONTINUOUS DISTRIBUTION

Cumulative Distribution function:

$$F_X(x) = P(X \leq x),$$

$F_X(x)$ is a continuous function of x ,

If $f(x)$ for $x \in (a, b)$ is a PDF, then

$$F(X) = \begin{cases} 0 & x \leq a \\ \int_a^x f(t)dt & a < x < b \\ 1 & b \leq x \end{cases}$$

$$f(x) = \frac{d}{dx} F(x).$$

EXAMPLE

If $f(x) = \frac{x}{2}, x \in (0,2)$ Find the CDF:

$$F(x) = \begin{cases} 0 & x \leq 0 \\ \int_0^x \frac{t}{2} dt = \frac{x^2}{4} & 0 < x < 2 \\ 1 & 2 \leq x \end{cases}$$

$$F(x) = \begin{cases} 0 & x \leq 0 \\ \frac{x^2}{4} & 0 < x < 2 \\ 1 & 2 \leq x \end{cases}$$

EXAMPLE:

If the CDF as follows $F(x) =$

$$\begin{cases} 0 & x \leq 0 \\ x^3 & 0 < x < 1 \\ 1 & 1 \leq x \end{cases}$$

find the PDF

$$\text{The } f(x) = 3x^2 \quad 0 < x < 1$$

EXPECTED VALUES OF DISCRETE RANDOM VARIABLES

The mean, or expected value, of a discrete or continuous random variable is the long run average.

The expected value can be negative , positive or zero. It depends on the random variable values

$$\mu = E(x)$$

Expected values

of discrete random variable

The ***expectation*** of a **discrete random variable** X taking the values $x_1, x_2, \dots$ and with probability mass function $f(x)$:

$$E[X] = \sum_i x_i P(X = x_i) = \sum_i x_i f(x_i)$$

The ***expectation*** of a **continuous random variable** X taking the values $a < x < b$ and with probability density function :

$$E[X] = \int_a^b x f(x) dx$$

We also call $E[X]$ the *expected value* or *mean* or μ of X .

EXAMPLE: EXPECTED VALUE

The following probability distribution of ER arrivals: find the expected value!!

x	10	11	12	13	14
$P(x)$	.4	.2	.2	.1	.1

$$\sum_{i=1}^5 x_i f(x) = 10(.4) + 11(.2) + 12(.2) + 13(.1) + 14(.1) = 11.3$$

EXAMPLE

If $f(x) = x/2, x \in (0,2)$

Find the expected value.

$$\text{And } \int_0^2 x * \frac{x}{2} dx = \int_0^2 \frac{x^2}{2} dx = \left. \frac{x^3}{6} \right]_0^2 = \frac{8}{6}$$

EXAMPLE

Let X be the discrete random variable that takes the values 1, 2, 4, 8, and 16, each with probability $1/5$. Compute the expectation of X .

$$E[X] = \sum_i x_i P(X = x_i) = \sum_i x_i f(x_i)$$

$$E[X] = 1 \cdot \frac{1}{5} + 2 \cdot \frac{1}{5} + 4 \cdot \frac{1}{5} + 8 \cdot \frac{1}{5} + 16 \cdot \frac{1}{5} = \frac{31}{5} = 6.2.$$

EXAMPLE

Let $f(x) = 1/3$ for $x \in (1, 4)$ then

$$E[X] = \int_a^b x f(x) dx$$

$$E(X) = \int_1^4 x \frac{1}{3} dx = \left[\frac{1}{6} x^2 \right]_1^4 = \frac{1}{6} (4^2 - 1^2) = \frac{5}{2}$$

THEOREM. Let X be a random variable, and let $g : \mathbb{R} \rightarrow \mathbb{R}$ be a function.

If X is discrete, taking the values $x_1, x_2, \dots$, then

$$E[g(X)] = \sum_i g(x_i) f(x_i)$$

If X is continuous, with probability density function f , then

$$E[g(X)] = \int_a^b g(x) f(x) dx$$

This formula will help us to calculate variance

EXAMPLE

Let X be the discrete random variable that takes the values 1, 2, 4, 8, and 16, each with probability $1/5$. Compute the expectation of $2X - 1$.

$$E[g(X)] = \sum_i g(x_i) f(x_i)$$

$$E[2X - 1] = \sum (2x - 1) f(x) = 1 * \frac{1}{5} + 3 * \frac{1}{5} + 7 * \frac{1}{5} + 15 * \frac{1}{5} + 31 * \frac{1}{5} = 11.4$$

VARIANCE

variance measures how far a set of numbers are spread out. A variance of zero indicates that all the values are identical. Variance is always non-negative: a small variance indicates that the data points tend to be very close to the Expected value and hence to each other, while a high variance indicates that the data points are very spread out around the mean and from each other.

The ***variance*** $Var(X)$ of a random variable X is

$$Var(X) = E[(X - \mu)^2] = E(X^2) - E(X)^2$$

**Standard
deviation:**

$$std = \sqrt{Var(X)}$$

EXAMPLE

Let X have $f(x) = 0.3^x \times 0.7^{1-x}$ $x=0,1$

x	0	1
$f(x)$	0.7	0.3

$$E(X) = \sum_x xP(x) = (0)(0.7) + (1)(0.3) = 0.3$$

$$E(X^2) = \sum_x (x)^2 P(x) = (0)^2(0.7) + (1)^2(0.3) = 0.3$$

$$VAR(X) = E(X^2) - E(X)^2 = 0.3 - 0.09 = 0.21$$

Example

Let X takes the values 2, 3, and 4 with probabilities 0.1, 0.7, and 0.2 respectively.

We can compute that $E[X] = 3.1$. then the variance=

$$\text{Var}(X) = E[X^2] - E[X]^2$$

$$\begin{aligned}\text{Var}(X) &= E[X^2] - (3.1)^2 = 0.1 \cdot 2^2 + 0.7 \cdot 3^2 + 0.2 \cdot 4^2 - 9.61 \\ &= 0.1 \cdot 4 + 0.7 \cdot 9 + 0.2 \cdot 16 - 9.61 \\ &= 0.4 + 6.3 + 3.2 - 9.61 \\ &= 0.29.\end{aligned}$$

EXAMPLE

Let $f(x) = 2x$ $0 < x < 1$, find expected value

And variance

$$E(x) = \int_0^1 2x^2 dx = 2/3$$

$$E(x^2) = \int_0^1 2x^3 dx = 0.5$$

$$\text{Var}(x) = 0.5 - (2/3)^2 = 0.056$$

PROPERTIES OF EXPECTED VALUES AND VARIANCES

If a and b are constants , X and Y are random variables then

■ $E(a) = a$

■ $E(ax) = aE(x)$

■ $E(x \pm y) = E(x) \pm E(y)$

■ $E(ax \pm by) = aE(x) \pm bE(y)$

PROPERTIES OF EXPECTED VALUES AND VARIANCES

**Example, if $E(X)=5$ and $E(Y)=6$,
then**

$$E(17) = 17$$

$$E(X + 5) = 5 + 5 = 10$$

$$E(2X + 5) = 2 * 5 + 5 = 15$$

$$E(3X - 2Y) = 3 * 5 - 2 * 6 = 3$$

PROPERTIES OF EXPECTED VALUES AND VARIANCES

If a and b are constants , X and Y are independent random variables then

■ $var(a) = 0$

■ $var(ax) = a^2 var(x)$

■ $var(x \pm y) = var(x) + var(y)$

■ $var(ax \pm by) = a^2 var(x) + b^2 var(y)$

PROPERTIES OF EXPECTED VALUES AND VARIANCES

Example: *if $Var(X) = 4$ and $Var(Y) = 9$, X and Y are independent.*

- $Var(2X + 3Y) = 4 * Var(X) + 9 * Var(Y)$
 $= 4 * 4 + 9 * 9 = 97$
- $Var(2X - 3Y) = 4 * Var(X) + 9 * Var(Y)$
 $= 4 * 4 + 9 * 9 = 97$
- $Var(-19) = 0$
- $Var(-12X + 14) = 144 * 4 = 576$